Name = M Kazim Seat NO#2312202

OL	Temp	Hd	W	R
Sunny	Hot	High	Weak	NO
Sunny	mild	High	strong	NO
cloudy	Hot	High	Weak	Yes
Rainy	mild	High	Weak	Yes
Rainy	cool	Normal	Weak	Yes
Rainy	cool	Normal	Strong	NO
cloudy	mild	Normal	strong	Yes
Sunny	cool	Normal	Weak	No

→ Entropy of (R)

Total records = 8

Yes = 4

No = 4

$$H(S) = -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$= 0.5 + 0.5$$

$$= 1$$

⇒

Temperature	Total	Yes	No	entropy
HOT	2	1	1	1
Mild	3	2	1	1.35
cool	3	1	2	1.35

- Hot → $H(s) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$

Mild → $H(s) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 1.1396 + 0.21 =$
$= 1.35$

cool → $H(s) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 1.35$

- Weight entropy (T) $= \frac{2}{8}(1) + \frac{3}{8}(1.35) + \frac{3}{8}(1.35)$
$= 0.805$

- IG (Hot) = H(S) − weight entropy
$= 1 - 0.805$
$= 0.194$

⇒

OL	Total	Yes	No	entropy
Sunny	3	0	3	0
cloudy	2	2	0	0
Rainy	3	2	1	1.3499

Rainy → $(H(s) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 1.3493$

- Weight entropy $= \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(1.3499)$

  $= 0.506$

- $IG(OL) = H(s) -$ Weight entropy

  $= 1 - 0.506$

  $= 0.4937$

⇒

Humidity	Total	Yes	No	entropy
High	4	2	2	1
Normal	4	2	2	1

High → $H(s) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1$

Normal → $H(s) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1$

- Weight entropy $= \frac{4}{8}(1) + \frac{4}{8}(1)$

  $= 1$

- IG (Humidity) $= H(s) -$ weight entropy

  $= 1 - 1$

  $= 0$

⇒ W	Total	Yes	No	entropy
Weak	5	3	2	1.1167
Strong	3	1	2	1.349

Weak ⇒ $H(S) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 1.1167$

Strong ⇒ $H(S) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 1.349$

Weight entropy $= \frac{5}{8}(1.1167) + \frac{3}{8}(1.349)$

$= 0.6979 + 0.5058 = 1.203$

*

$IG(W) = H(S) -$ weight entropy

$= 1 - 1.203$

$= -0.2037$

Outlook has maximum info gain value so it is taken as root node